31-3-26

TUTORIAL-10
(Boosting)

Attribute / Feature Age	Income	w_i	credit Score	w_i^{new}	Target (Loan approved)	Pred with w_i^{new}
25	40	0.2	650	0.14	Yes → +1	0.125 +1
30	60	0.2	700	0.14	Yes → +1	0.125 +1
45	35	0.2	600	0.14	No → −1	0.125 −1
50	30	0.2	580	0.14	No → −1	0.125 −1
28	55	0.2	720	0.4	Yes → +1 / No → −1	0.5 +1

decision stump.
↓ Many
AdaBoost

1) Initialize weights: $w_i = \frac{1}{N} = \frac{1}{5} = 0.2$

2). decision stump

Age < 40 → Yes (+1)
≥ 40 → No (−1)

3) weighted Error

$$E_t = \sum_{i=1}^{N} w_i \cdot \underbrace{I}_{\text{Indicator function}}(\underbrace{y_i}_{\text{Actual}} \neq \underbrace{h_t(x_i)}_{\text{predic}})$$

$= [0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0] = 0.$

classifier wt. $0.2 \times 1 = 0.2$

4). Model error.

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1 - E_t}{E_t}\right) = \frac{1}{2} \ln\left(\frac{1-0}{0}\right) \Rightarrow \infty$$

$$= \frac{1}{2} \ln\left(\frac{1-0.2}{0.2}\right) = 0.693$$

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5). Update weights:

$$w_i^{new} \rightarrow w_i e^{-y_i \alpha_1 h_1(x_1)}.$$

$$\downarrow$$

$$e^{-\infty} = 0.$$

6) prediction

Model will stop training (x_1)

$$H(x) = h_1(x_1)$$

(if misclassified)

$$0.2 \times e^{-[1 \times 0.693 \times 1]}$$

$$0.2 \times e^{-[1 \times 0.693 \times -1]}.$$

Normalized wts. $= \frac{w_i}{\sum w_i} = \frac{0.1}{0.8} = 0.125 \rightarrow$ first 4

$0.2 \rightarrow 0.125$
(wts decreasing)

$\frac{0.4}{0.8} = 0.5 \rightarrow$ last one

$0.2 \rightarrow 0.5$
(increasing).

7) Final Prediction.

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x_i)\right)$$

$$= \text{sign}[\alpha_1 h_1(x_i) + \alpha_2 h_2(x_2) + \dots].$$

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